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Comparative Analysis of Missing Value Imputation Methods with a Hybrid Approach for HCV Prediction Using Machine Learning

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Overview Of the Talk

Background and Motivation of the Study

Problem Statement and Research Gap

Objectives and Key Contributions

Dataset Description and Feature Overview

Methodology and Proposed Approach

Experimental Results and Performance Analysis

Conclusion and Future Research Directions

Background (HCV Overview)

- ❑ Hepatitis C Virus (HCV) affects about 58 million people worldwide and is a major global health concern.
- ❑ It can lead to serious liver diseases like cirrhosis and cancer, making early detection very important.

Problem Context

- ❑ Clinical data often has missing values from errors or incomplete tests.
- ❑ This reduces model accuracy, so handling missing data is important.



Research Gap

- ❖ Most studies use complete data, but real clinical data is often incomplete.
So, better methods are needed to handle missing data.

Key Contributions

- Proposed a hybrid method for handling missing data.
- Evaluated multiple ML models for prediction.
- Tested model stability under different conditions.

Research Objectives

- ❑ Study how missing data affects clinical data and model performance.
- ❑ Use a mix of MICE and median methods to fill missing values.
- ❑ Test different models under various missing conditions and predict Hepatitis C progression.

Dataset

Name	Meaning
age	Age of the patient
ALB	Albumin level in the blood
ALP	Alkaline phosphatase level in the blood
ALT	Alkaline aminotransferase level in the blood
AST	Aspartate aminotransferase level in the blood
BIL	Bilirubin level in the blood
CHE	Choline esterase level in the blood
CHOL	cholesterol level in the blood
CREA	Serum creatinine level in the blood
GGT	Gamma-glutamyl-transferase level in the blood
PROT	Total protein level in the blood
sex	Sex of the patient
[target]binary condition	0: healthy control
	1: hepatitis C
[target]disease category	0: healthy control
	1: hepatitis C
	2: hepatitis C, and fibrosis
	3: hepatitis C, fibrosis, and cirrhosis

Table 1: Understanding and evaluation of clinical attributes. IU: international units. L: liter. mg: milligram. g: gram. dl: deciliter.

Target Variables & Data Challenges

- ❑ The study uses binary and multiclass targets for HCV.
- ❑ Binary shows healthy vs infected, while multiclass shows disease stages.
- ❑ The dataset is imbalanced, so training is harder.

Missing Data Summary

Variable	Missing values, n	Variable type
ALB	1	Numeric biochemical predictor
ALP	18	Numeric biochemical predictor
ALT	1	Numeric biochemical predictor
CHOL	10	Numeric biochemical predictor
PROT	1	Numeric biochemical predictor
Total	31	

- ❑ The dataset has 31 missing values, mainly in ALP and CHOL.
- ❑ This highlights the need for strong imputation methods.

Table 2: Summary of Missing Values Across Biochemical Variables

Complete Case Analysis (Baseline)

CCA removes rows with missing values.

So, 589 of 615 samples were used.

It is used as a baseline to compare other methods.

Imputation Methods

- **MICE (Multiple Imputation by Chained Equations)**
 - MICE fills missing data using variable relationships.
- **Median Imputation**
 - Median replaces missing values with the middle value.
- **Sampling Imputation**
 - Sampling fills values from existing data randomly.
- **Key Insight:**
 - Each method affects model performance differently.

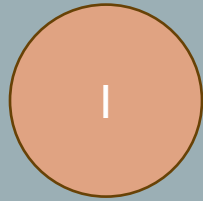
Proposed Hybrid Approach

Improved Imputation Strategy (MICE + Median)

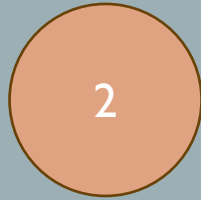
Proposed Approach:

- A hybrid method combines MICE and median imputation.
- Median stabilizes data, while MICE improves accuracy.
- This gives more reliable and stable results.

Leakage-Safe Preprocessing Pipeline



80:20 Split



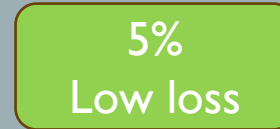
Fit Imputer



Scale and Fit

- ❑ Data is split (80:20) before processing to avoid data leakage.
- ❑ Training data is used for scaling and missing values, with balanced class distribution maintained.

Artificial MCAR Experiment Setup



- ❑ Missing values are applied only to numeric features, excluding target and ID columns.
- ❑ Model performance is evaluated only on the artificially masked data.

Regression performance under complete-case analysis

Method	Model	R^2	MAE	RMSE
Baseline	Mean Target	-0.011± 0.021	0.452±0.037	0.757±0.095
CCA	Decision Tree	0.642±0.184	0.115±0.030	0.387±0.082
CCA	Lasso Regression	0.655±0.153	0.199±0.025	0.385±0.072
CCA	Linear Regression	0.654±0.161	0.206±0.025	0.385±0.071
CCA	Linear SVR	0.655±0.145	0.172±0.029	0.388±0.080
CCA	MLP Regressor	0.728±0.156	0.153±0.029	0.342±0.089
CCA	RBF-SVR	0.699±0.070	0.173±0.027	0.365±0.064
CCA	Random Forest	0.798±0.123	0.088±0.021	0.288±0.061
CCA	XGBoost	0.827±0.102	0.088±0.021	0.270±0.060

Ensemble models, especially XGBoost, performed best for prediction, while simpler models performed worse.

Table 3: Regression performance of machine learning models under complete-case analysis

Classification performance under complete-case analysis

Method	Model	MCC	FI Score	Accuracy	ROC-AUC
Baseline	Majority Class	0.000±0.000	0.000±0.000	0.878±0.000	0.500±0.000
CCA	Linear SVM	0.911±0.064	0.917±0.059	0.984±0.012	0.980±0.028
CCA	XGBoost	0.914±0.057	0.920±0.053	0.984±0.011	0.998 ± 0.002

CCA with advanced models like XGBoost gives very high and stable classification performance.

Table 4: Top Classification Performance Comparison (Baseline vs CCA Models)

Overall MICE Performance

CCA with models like XGBoost gives accurate and stable results.

Missing Rate (%)	MAE(Mean±SD)	RMSE (Mean±SD)	MSE(Mean±SD)	RMSE/SD(Mean ± SD)
5%	10.5504±1.1958	26.011±10.5226	781.7587±752.1219	0.9215±0.1639
10%	10.3532±0.7308	24.5062±5.4655	628.9331±329.7781	0.9319±0.1052
20%	11.6139±1.2831	29.7098±6.4496	922.1889±388.9674	1.0171±0.1046

Table 5: Imputation Performance of MICE under MCAR Missingness

Feature-wise Imputation Performance at Different Missing Rates (MICE)

Missing Rate(%)	Feature	MAE (Mean±SD)	RMSE (Mean±SD)	MSE (Mean±SD)	RMSE/SD (Mean±SD)
5%	ALB	3.247±0.575	4.174±0.975	18.323±9.172	0.722±0.169
5%	CHE	1.402±0.206	1.867± 0.374	3.618±1.515	0.846±0.169
5%	CHOL	0.767±0.094	0.985±0.142	0.990±0.312	0.870±0.126
10%	Age	8.134±0.846	9.937±0.885	99.848±18.080	0.988±0.088
10%	CHE	1.414±0.181	1.869±0.298	3.577±1.187	0.847±0.135
10%	CHOL	0.810±0.064	1.042±0.094	1.094±0.201	0.920±0.083
20%	CHE	1.478±0.108	1.943±0.187	3.810±0.753	0.881±0.085
20%	CHOL	0.817±0.092	1.089±0.169	1.214±0.421	0.962±0.149
20%	PROT	3.424±0.334	4.720±0.557	22.572±5.203	0.874±0.103

CHOL and CHE perform best, while errors increase slightly with more missing data but remain stable overall.

Table 6: Feature-wise Imputation Performance of MICE under MCAR Artificial Missingness

Overall Imputation Performance Stability Across Missing Rates(Median)

Missing (%)	Rate	MAE(Mean±SD)	RMSE (Mean±SD)	MSE(Mean±SD)	RMSE/SD(Mean ± SD)
5%		10.398±1.694	29.462±12.293	1011.567±889.313	1.025±0.169
10%		9.823±0.846	26.820±6.348	757.576±385.775	1.024±0.125
20%		10.050±0.668	29.908±6.638	936.321±399.830	1.024±0.100

Performance is best at 10% missing data and remains stable with only a slight increase at 20%.

Table 7: **Median imputation performance under MCAR missingness**

Feature-wise Imputation Performance at Different Missing Rates (Median)

Missing Rate(%)	Feature	MAE (Mean $\pm$ SD)	RMSE (Mean $\pm$ SD)	MSE (Mean $\pm$ SD)	RMSE/SD (Mean $\pm$ SD)
5%	ALB	4.125 $\pm$ 0.643	5.545 $\pm$ 1.132	31.962 $\pm$ 12.814	0.959 $\pm$ 0.196
5%	CHE	1.714 $\pm$ 0.249	2.249 $\pm$ 0.373	5.189 $\pm$ 1.700	1.020 $\pm$ 0.169
5%	CHOL	0.883 $\pm$ 0.108	1.137 $\pm$ 0.176	1.322 $\pm$ 0.419	1.004 $\pm$ 0.155
10%	CHE	1.664 $\pm$ 0.205	2.160 $\pm$ 0.292	4.746 $\pm$ 1.297	0.979 $\pm$ 0.133
10%	CHOL	0.934 $\pm$ 0.091	1.182 $\pm$ 0.124	1.412 $\pm$ 0.295	1.044 $\pm$ 0.110
10%	PROT	3.769 $\pm$ 0.427	5.249 $\pm$ 0.773	28.116 $\pm$ 8.516	0.971 $\pm$ 0.143
20%	ALB	4.161 $\pm$ 0.353	5.824 $\pm$ 0.785	34.499 $\pm$ 9.185	1.007 $\pm$ 0.136
20%	CHE	1.703 $\pm$ 0.141	2.219 $\pm$ 0.170	4.949 $\pm$ 0.755	1.

Best at 10%, and still stable at 20% with slight change.

Table 8 : Feature-wise Imputation Performance of Median under MCAR Artificial Missingness

Overall Hybrid Imputation Performance

Missing Rate (%)	MAE(Mean ± SD)	RMSE (Mean±SD)	MSE(Mean±SD)	RMSE/SD(Mean±SD)
5%	9.838±1.408	26.493±11.620	830.148±810.792	0.923±0.167
10%	9.391±0.695	24.212±5.912	619.435 ± 346.817	0.936 ± 0.116
20%	9.933±0.686	27.879±6.909	822.553±395.476	0.955±0.099

Imputation stays stable across all missing rates, with best performance at 10%.

Table 9: Imputation Performance of Hybrid Imputation under MCAR Missingness

Feature-wise Imputation Performance at Different Missing Rates (Hybrid)

Missing Rate(%)	Feature	MAE (Mean±SD)	RMSE (Mean±SD)	MSE (Mean±SD)	RMSE/SD (Mean±SD)
5%	CHE	1.492±0.218	1.952±0.343	3.922±1.392	0.885±0.155
5%	CHOL	0.804±0.098	1.032±0.156	1.088±0.347	0.911±0.138
5%	PROT	3.416±0.599	4.684±0.880	22.674±8.263	0.867±0.163
10%	CHE	1.488±0.189	1.919±0.261	3.749±1.036	0.870±0.118
10%	CHOL	0.847±0.072	1.077±0.105	1.170±0.230	0.951 ± 0.093
10%	PROT	3.339±0.418	4.559±0.740	21.307±7.025	0.844±0.137
20%	CHE	1.518±0.115	1.957±0.148	3.852±0.587	0.887±0.067
20%	CHOL	0.809±0.064	1.052±0.087	1.115±0.181	0.929±0.077
20%	PROT	3.441±0.311	4.715±0.428	22.404±4.040	0.873±0.079

Imputation is stable across all rates, with CHOL and CHE performing best and PROT having higher error.

Table 10: Feature-wise Imputation Performance of Hybrid under MCAR Artificial Missingness

Overall Sampling Performance

Missing Rate (%)	MAE(Mean ± SD)	RMSE (Mean±SD)	MSE(Mean±SD)	RMSE/SD(Mean ± SD)
5%	15.682±2.090	39.078±10.863	1639.205±996.343	1.389±0.159
10%	15.023±1.141	36.066±5.498	1329.446±418.173	1.401±0.100
20%	15.226±0.929	40.451±5.362	1663.604±441.233	1.420±0.104

Imputation is consistent but has higher errors and slightly lower stability.

Table 11: Imputation Performance of sampling under MCAR Missingness

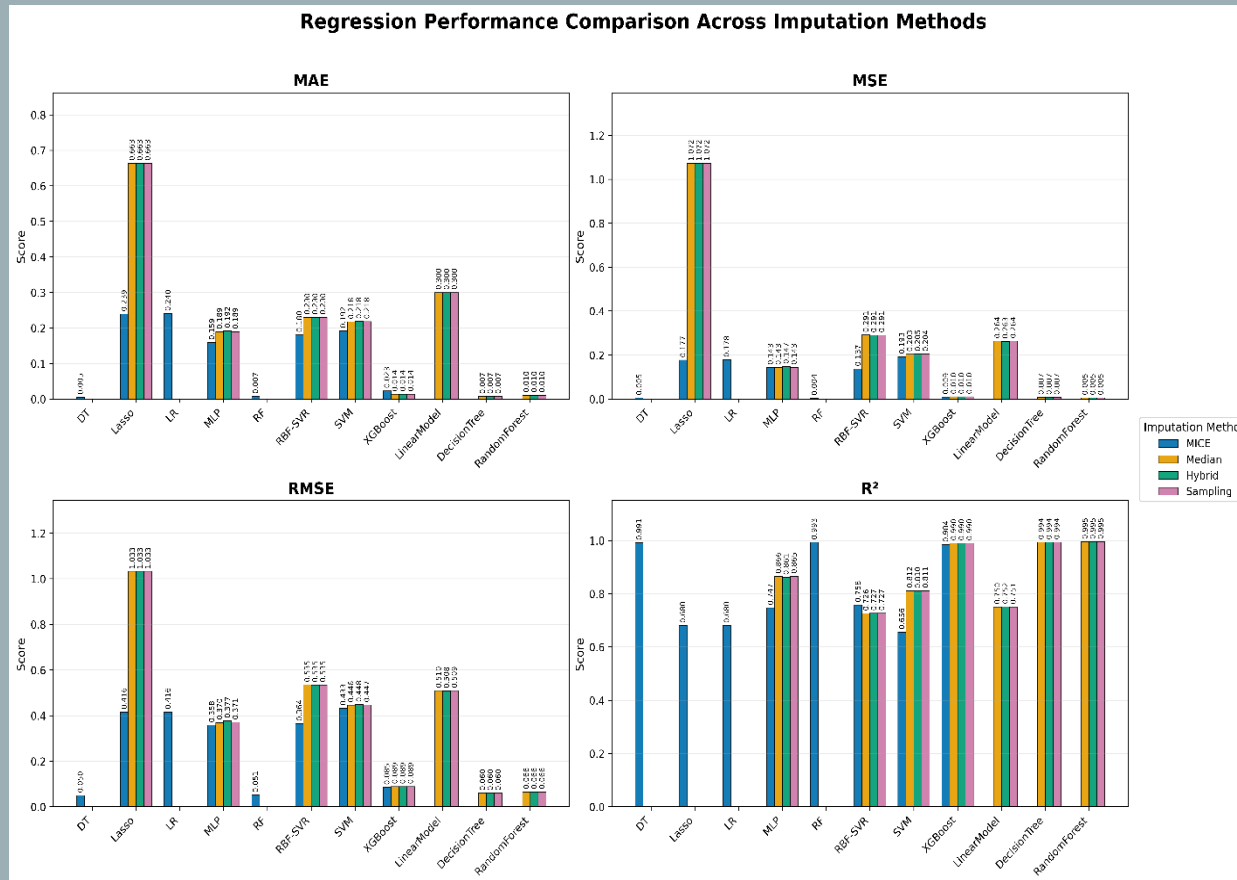
Feature-wise Imputation Performance at Different Missing Rates (Sampling)

Missing Rate(%)	Feature	MAE (Mean±SD)	RMSE (Mean±SD)	MSE (Mean±SD)	RMSE/SD (Mean±SD)
5%	CHE	2.495±0.303	3.163±0.400	10.155±2.576	1.434±0.181
5%	CHOL	1.223±0.142	1.556±0.214	2.466±0.686	1.374±0.189
5%	PROT	5.490±0.727	7.109±0.900	51.306±12.867	1.316±0.167
10%	CHE	2.396±0.253	3.046±0.338	9.387±2.074	1.381±0.153
10%	CHOL	1.306±0.193	1.649±0.156	2.741±0.505	1.455±0.138
10%	PROT	5.756±0.644	7.593±1.059	58.712±16.477	1.405±0.196
20%	CHE	2.418±0.201	3.112±0.288	9.763±1.855	1.411±0.131
20%	CHOL	1.259±0.081	1.607±0.091	2.589±0.290	1.418±0.081
20%	PROT	5.690±0.352	7.625±0.627	58.515±9.578	1.411±0.116

Imputation is stable across all rates, with CHOL best and PROT hardest to predict.

Table 12: Feature-wise Imputation Performance of Sampling under MCAR Artificial Missingness

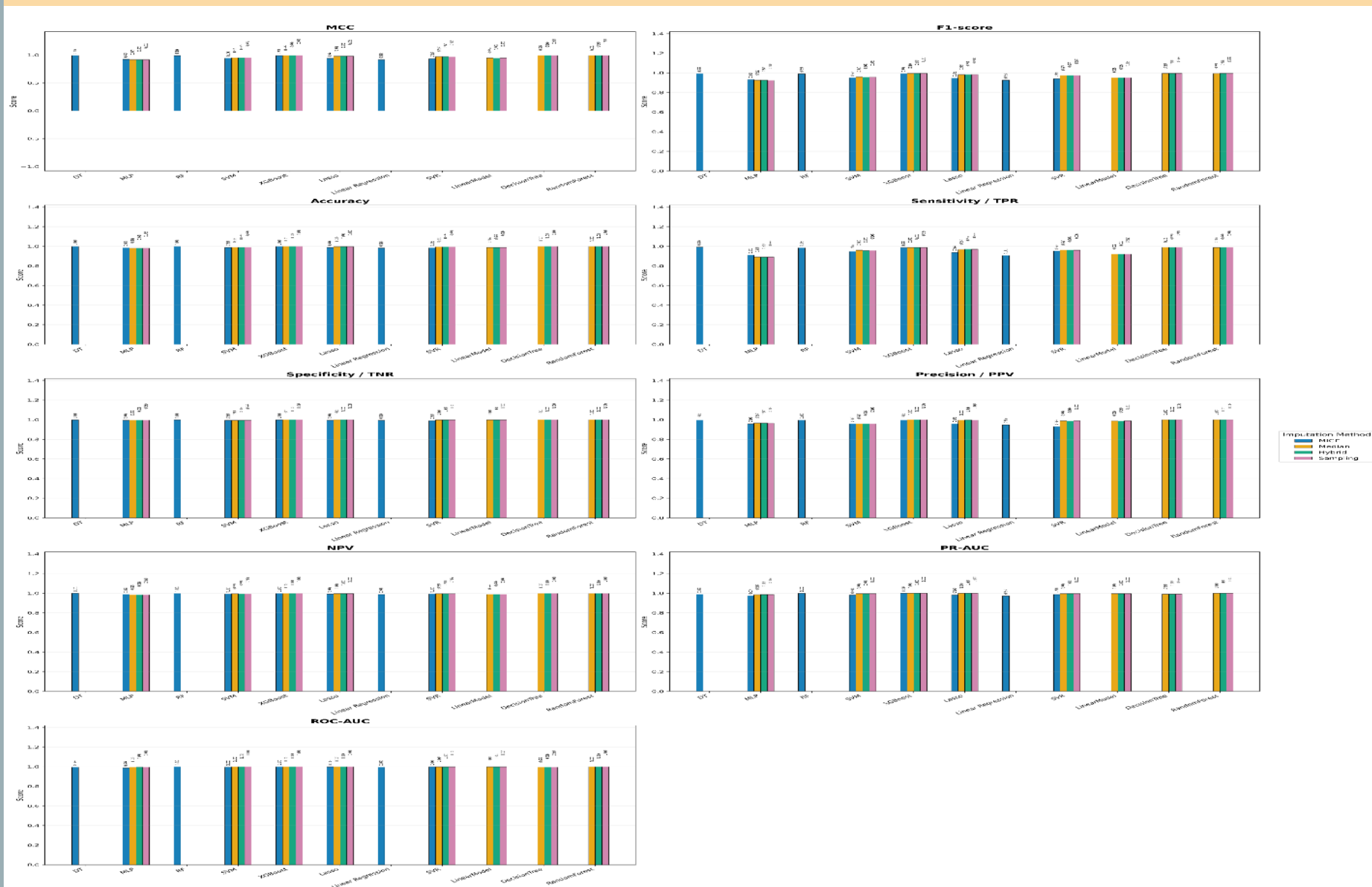
Regression Performance Comparison



Ensemble models consistently outperform others, showing lower errors and higher prediction accuracy.

Figure 1: Comparative Analysis of Regression Performance Across Imputation Methods

Classification Performance Comparison



Ensemble models, particularly XGBoost, consistently achieve the highest performance across all evaluation metrics and imputation methods.

Figure 2 : Comparative Evaluation of Classification Performance Across Imputation Methods

Conclusion

- Ensemble models (especially XGBoost) give the best and most accurate results.
- Handling missing data has a big impact on model performance.
- Hybrid MICE + Median method provides stable and reliable imputation.
- Better preprocessing and model choice improve overall prediction accuracy.

Limitations

- ☐ The study uses only one dataset, so results may not apply everywhere.
- ☐ Class imbalance and simulated missing data may affect accuracy.
- ☐ No external dataset was used for validation.

Future Work

- Use larger real-world datasets and test with realistic missing data.
- Improve model explainability and integrate it into clinical systems.

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Thank you

Q & A

I would be happy
to take any
questions.





THE END